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Weifang Delco Chemical Co., Ltd. 潍坊达科化工有限公司

Room616,Taihualingyu Building, Shengli East Street, Kuiwen District, Weifang, Shandong, China
yulan@delcochem.com Tel: 0086-536-8567610

PRODUCT SPECIFICATION

Product Name：UREA 46% MIN.

Items	Specifications
Appearance	White Crystal 0.85~2.8mm
Nitrogen	>46.0%
Biuret	≤1.0%
Moisture	≤0.5%
Fe	≤0.5%
Reductive Matter (as H3PO3)	≤120 mg/kg
Insoluble	≤0.04%
Urea Purity	>99.0%
Conclusion: The product complies with the specification in-house Urea Standard.	



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MATERIAL SAFETY DATA SHEET

Supplier Information
Manufacturer: WEIFANG DELCO CHEMICAL CO.,LTD
Room616,Taihualingyu Building, Shengli East Street, Kuiwen District, Weifang, Shandong, China
Telephone (86) 536-8567610
Fax (86) 536-8867610
Emergency Telephone (86) 15653678177 (24 hours)

Section 1 Product Identification

Product Identifier: Urea **CAS#:** 57-13-6
Formula: CH_4N_2O

Section 2 Hazardous Ingredients

None according to the Controlled Products Regulations.

Chemical Name	CAS #	Weight %	ACGIH TLV
Urea	57-13-6	100%	none

Section 3 Hazard Identification

Emergency overview: Urea is a minor irritant to humans.
Primary Routes of Entry: Ingestion, Skin Contact, Eye Contact, and Inhalation

Section 4 First Aid Measure

FIRST AID PROCEDURES WHEN:
In Contact with the Eyes: Immediately flush eyes thoroughly for 15 minutes with running water. Hold eyelids open during flushing. If irritation persists, repeat flushing and obtain medical attention.
In Contact with the Skin: Remove contaminated clothing. Wash affected areas thoroughly with soap and water. If irritation persists, obtain medical advice.
Inhaled: Move victim to fresh air.
Ingested: If victim is alert and not convulsing, rinse mouth out and give several glasses of water to dilute material. Induce vomiting. If spontaneous vomiting occurs, have victim lean forward with head down to avoid breathing in of vomitus. Rinse mouth and administer more water. Obtain medical attention.

Section 5. Fire Fighting Measures

Flash Point: Non flammable
Auto Ignition Temperature: Not combustible
Flammability Limits in Air (%): UEL: Not applicable, LEL: Not applicable
Extinguishing Media: Use appropriate media for surrounding fire
Special Fire Fighting Procedures: A self-contained breathing apparatus should be worn in a fire involving the material. Toxic gas and vapor (carbon monoxide, ammonia, nitrogen oxide) may be released by decomposition.
Other Fire/Explosion Hazards: Reacts with Nitrates with risk of fire and explosion. Not combustible, but urea decomposes at > 133°C generating toxic NO_x fumes and ammonia gas.3

Section 6 Accidental Release Measures

Collect all spilled material and store in containers suitable for recycle or disposal.

Section 7 Handling and Storage

Urea is extremely hygroscopic at relative humidity above approximately 70%. Store in a



cool, wellventilated area. Keep away from heat, sparks, and flame. Keep containers closed. Do not expose sealed containers to temperatures above 49°C. Keep dry.

Section 8 Exposure Controls/personal Protection

Ventilation: Natural ventilation or local exhaust systems are adequate for ventilation.
Eye Protection: Use chemical safety goggles when there is potential for eye contact.
Respiratory Protection: A dust mask should be suitable for most applications. Wear respiratory protection for dust/mist where ventilation is inadequate.
Skin Protection: Cotton gloves and protective clothing should be used.
Other Precautions: Locate safety shower and eyewash station close to chemical handling area.
EXPOSURE LIMITS:
OSHA PEL: 15 mg/m3 total dust 5 mg/m3 respirable dust

Section 9 Physical and Chemical Properties

Physical State: Solid @ 20°C and 101 Kpa, crystal or white powder
Appearance and Odour: White granules, odorless or slight ammonia odor
Odor Threshold: Not applicable
Boiling Point: 135°C at 101 kPa (Decomposes)
Freezing Point: 133°C
Specific Gravity: Solid at 20°C = 1.335 t/m 3
Vapour Pressure: 80 Pa at 20°C
Bulk Density: Granules: 750 kg/m3 (48 lb/cu ft)
Solubility (Water): 78 g/100 ml (5°C); 119.3 g/100 ml (25°C)
pH: ≈7.5 at 100g/L
Coefficient of Water/Oil
Distribution: -1.59 at 20-25°C
Floatability (Water): Sinks and mixes
Molecular Weight: 60.07
Hygroscopicy: @ > 70% Rel. Humidity

Section 10 Stability and Reactivity Data

Stability: Stable under normal conditions. Decomposes under fire conditions.
Incompatibility (Materials to Avoid): Avoid contact with caustics and akalines. Mixing with phosphate or other nitrate fertilizers will increase the hygroscopic nature of urea.
Hazardous Products
Of Decomposition: Decomposes in flame or on hot surface, giving off toxic nitrous vapors. Decomposes when heated, giving off ammonia vapor.
Hydrolyzes slowly on contact with air. Soluble in water.

Section 11 Toxicological Information

Recommended Exposure Limit: None established for this product
Toxicological Data (Acute): LD50 (oral, rat): 14300-15000 mg/kg
LD50 (oral, mouse): 11500-13000 mg/kg
Toxicological Data (Chronic): NOAEL (oral, mouse): 6750 mg/kg
NOAEL (oral, rat): 2250 mg/kg
(NOAEL: No Observable Adverse Effect Level)
Carcinogenicity Data: The ingredient(s) of this product is (are) not classified as carcinogenic by ACGIH (American Conference of Government Industrial Hygienists) or IARC (Internal Agency for Research

on Cancer), not regulated as carcinogens by OSHA (Occupational Safety & Health Association), and not listed as carcinogens by NTP (National Toxicology Program).
Reproductive Effects: No reproductive effects noted.
Mutagenicity Data: Not mutagenic.
Teratogenicity Data: Not teratogenic.
EFFECTS OF EXPOSURE WHEN:
Inhaled: Dust is irritating to the nose, throat, and respiratory tract, and may cause coughing and sneezing.
In Contact with the Skin: Prolonged and repeated contact may cause mild irritation. May be irritating at >10% concentrations.
In Contact with the Eyes: May cause irritation, redness, and pain.
Ingested: Ingestion of large amounts of dust may cause gastrointestinal upset and abdominal pain.
Other Health Effects: Forms corrosive solutions when dissolved in water (or tears). High blood concentrations of urea increase the risk of glaucoma.

Section 12 Ecological Information

Toxicological Data (Acute): LC_{50} (fish, 96 hr): >9,100 mg/L
EC_{50} (daphnia, 24 hr): >10,000 mg/L
Do not contaminate bodies of water with urea.

Section 13 Disposal Considerations

Consult local, provincial and federal environmental regulatory agencies for disposal procedures and sites.

Section 14 Transport Information

Controlled Products Regulations Classifications.....Not regulated
Identification Number.....Not required
OSHA Hazard Communication (29CFR 1910.1200) Classification.....Not regulated
Canadian TDG Act Shipping Description Shipping Name.....Urea
U.S. DOT Classification.....Not regulated

Section 15 Regulatory Information

This product is listed on the Canadian Environmental Protection Act domestic substances list. Product is not regulated by Transportation of Dangerous Goods or WHMIS

Section 16 Preparation Information

NOTICE
Seller warrants that this product conforms to the specifications stated herein. Buyer assumes all risks associated with its use, mixing, blending, treatment, storage, disposal, transportation, and handling whether alone or in combination with other substances. Seller makes no other warranty of any kind, expressed or implied, and all other warranties, including warranties of quality, merchantability, and fitness for a particular use, are disclaimed. Seller's liability is limited to purchase price.

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Urea

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